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Exp No: 15	Analyse various types of servers using Webalizer tool

AIM:

Toanalyse the log files of various typesofservers (web, proxy and FTP servers) using the Webalizer web server log analysis tool and to study theusage statistics generated by it.

ALGORITHM / PROCEDURE:

1. Install Webalizer in Linux using sudo apt install webalizer apache2.
2. Locate the log files of the servers to be analysed (Apache, Nginx, IIS, FTP, Squid) as given in the table below.
3. Edit the configuration file /etc/webalizer/webalizer.conf and set LogFile, LogType, OutputDir and HostName.
4. Run Webalizer for each server. The option -n gives the host name for the report, -o gives the output folder and -F gives the log type (clf, w3c, ftp, squid).
5. Open the generated usage_YYYYMM.html file in a web browser (http://localhost/webalizer/apache/) to see the report.
6. Study the monthly summary, the daily graph, the top URLs, the top clients, the user agents and the HTTP response codes.
7. Compare the results of thedifferent servers and note down the inference.

Servers and their log formats

Server type	Log file	Log format Common Log Format (CLF) CLF compatible (combined)	Webalizer option
Apache web server	/var/log/apache2/access.log		-F clf
Nginx web server	/var/log/nginx/access.log		-F clf
IIS web server	C:\inetpub\logs\LogFiles\W3SVC1\u_ex261008.log	W3C extended	-F w3c
FTP server (vsftpd)	/var/log/xferlog /var/log/squid/access.log	wu-ftpd xferlog	-F ftp
Squid proxy server		Squid native	-F squid

webalizer.conf (important lines)

```
LogFile         /var/log/apache2/access.log clf
LogType
OutputDir       /var/www/html/webalizer/apache
HostName        apache-server
PageType        htm*
PageType        php
PageType        cgi
HideSite        localhost
```

Sample log records

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```
Apache (CLF):
192.168.1.25 - - [08/Oct/2026:10:41:07 +0530] "GET /index.html HTTP/1.1" 200 4521
192.168.1.12 - - [08/Oct/2026:10:41:09 +0530] "GET /css/style.css HTTP/1.1" 304
10.0.2.2 - - [08/Oct/2026:10:42:15 +0530] "GET /missing.html HTTP/1.1" 404 492
IIS (W3C):
#Fields: date time s-ip cs-method cs-uri-stem sc-status sc-bytes
2026-10-08 05:12:44 192.168.1.30 GET /default.aspx 200 6120
```

DIAGRAM:

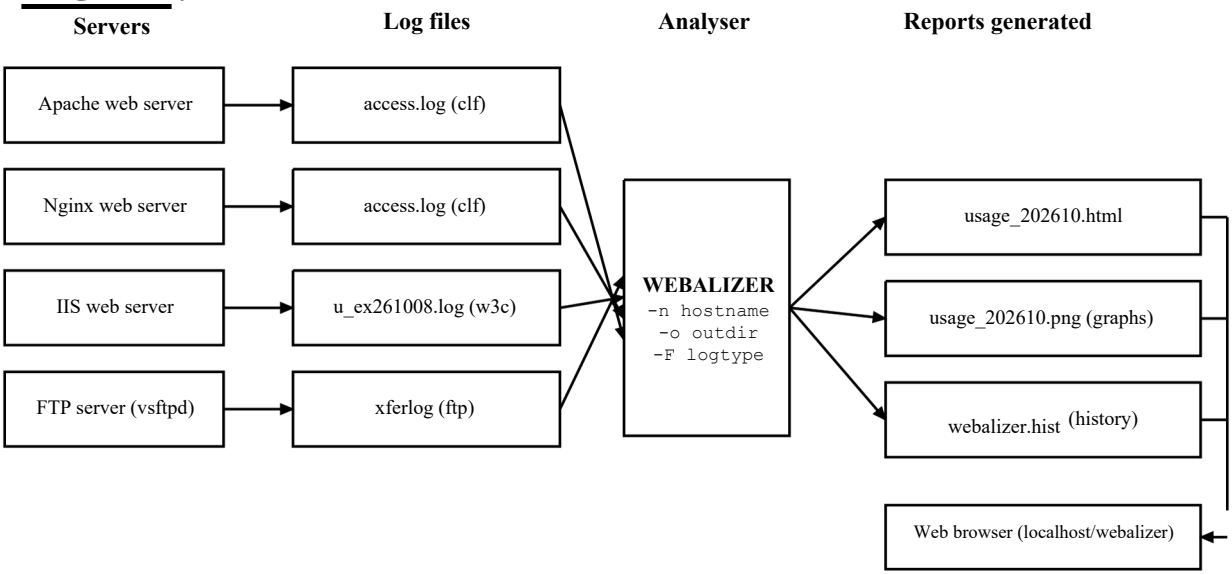


Fig 1: Log analysis of different servers using Webalizer

OUTPUT:

```
userUSER @userUSER -VirtualBox: ~

userUSER @userUSER -VirtualBox:~$ sudo webalizer -n apache-server -o /var/www/html/webalizer/apache -F clf
/va   r/log/apache2/access.log Webalizer V2.23-08 (Linux 5.15.0-91-generic) English
Using logfile /var/log/apache2/access.log (clf)
Creating output in /var/www/html/webalizer/apache
Hostname for reports is 'apache-server'
History file not found...
Generating report for October 2026
Generating summary report Saving history information... 1243 records in 0 seconds userUSER @userUSER -
VirtualBox:~$ sudo webalizer -n nginx-server -o /var/www/html/webalizer/nginx -F clf /var/
log/nginx/access.log Using logfile /var/log/nginx/access.log (clf)
Creating output in /var/www/html/webalizer/nginx
Generating report for October 2026
Generating summary report
Saving history information...
856 records in 0 seconds
userUSER@userUSER -VirtualBox:~$ sudo webalizer -n iis-server -o /var/www/html/webalizer/iis -F w3c u_ex26100
  8.log
Using logfile u_ex261008.log (w3c) Creating
output in /var/www/html/webalizer/iis
Generating report for October 2026
Generating summary report
Saving history information...
522 records in 0 seconds
USERuser @userUSER -VirtualBox:~$ sudo webalizer -n ftp-server -o /var/www/html/webalizer/ftp -F ftp
/var/log/   xferlog Using logfile /var/log/xferlog (ftp)
Generating report for October 2026
64 records in 0 seconds
```

Report of the Apache web server

Monthly Statistics for October 2026	
Total Hits	1243
Total Files	1131
Total Pages	402
Total Visits	138
Total KBytes	18642
Total Unique Sites	57
Avg / Max Hits per Day	155 / 276
Avg / Max Files per Day	141 / 252 50
Avg / Max Pages per Day	/ 93
Avg / Max Visits per Day	17 / 23 2330
Avg / Max KBytes per Day	/ 3342

Top URLs by hits

#	Hits	%	URL
1	214	17.2%	/index.html
2	188	15.1%	/css/style.css
3	163	13.1%	/images/logo.png
4	97	7.8%	/login.php
5	76	6.1%	/about.html
6	52	4.2%	/contact.html
7	41	3.3%	/favicon.ico

Top clients

#	Hits	%	Client (site)
1	311	25.0%	192.168.1.25
2	204	16.4%	192.168.1.12
3	142	11.4%	10.0.2.2
4	96	7.7%	192.168.1.30

Hits by response code

HTTP response code	Hits	%
200 - OK	1019	82.0%

304 - Not Modified	112	9.0%
404 - Not Found	74	6.0%
301 - Moved Permanently	21 9	1.7%
403 - Forbidden	8	0.7%
500 - Internal Server Error		0.6%

Comparison of the servers

Server	Records	Total hits	Visits	KBytes	Most requested URL
Apache	1243	1243	138	18642	/index.html
Nginx	856	856	97	12408	/api/status
IIS	522	522	61	7915	/default.aspx
FTP (vsftpd)	64	64	9	41230	/pub/notes.pdf

Inference

- TheApache server is the busiest server and its highest load was on 08-Oct-2026 (276 hits).
- About82% of the requests were successful (200) and 6% gave 404 errors, which means some links on the siteare broken.
- The home page /index.html is the most visited page. Image and CSS files form most of the hits, whereas only 402 hits are actual pages.
- The same tool can analyse all the servers because only the log type option (-F) changes.

RESULT:

Thus the logfiles of various types of servers (Apache, Nginx, IIS, FTP) were analysed using the Webalizer tool and theusage statistics such as hits, visits, top URLs, clients and response codes were studied.